

Chapter 1 Bundles

1.1 Bundles

What is bundle?

1.1.1 Definitions

Definition 1.1.1.1

- In most general case, a **bundle** over an object B in a category $\mathcal{C}$ is simply an object E of $\mathcal{C}$ equipped with a morphism in $\mathcal{C}$ from E to B

$$p : E \rightarrow B$$

And let $(x : U \rightarrow B)$ be a generalized element of B . (??), the **fibre** E_x of the bundle over x is the pullback is the pullback x^*E

$$\begin{array}{ccc} x^*E = U \times_B E & \longrightarrow & E \\ \downarrow & & \downarrow p \\ U & \xrightarrow{x} & B \end{array}$$

Here, $U = 1$.

Such bundles over B is a category. Called **slide category** $\mathcal{C}_{/B}$ of $\mathcal{C}$ over B .

- The morphism $E_1 \rightarrow E_2$ is a diagram such that the following diagram commutes

$$\begin{array}{ccc} E_1 & \xrightarrow{\quad} & E_2 \\ & \searrow & \swarrow \\ & B & \end{array}$$

Some of Bundles are listed as following:

- Fibre Bundle;
- Micro Bundle;
- Principle Bundle;
- Vector Bundle;
- Bundle with Connection;

Cross Sections

Definition 1.1.1.2 A **cross section** of a bundle (E, p, B) is a

Product Bundle

Stiefel Variety

Definition 1.1.1.3 ...

Grasmannian Variety

Definition 1.1.1.4 ...

1.1.2 Morphisms of Bundles

1.1.3 Product and Fibre Products

1.1.4 Restriction and Induced Bundles

1.1.5 Local Properties

1.2 Connections

Chapter 2 Fibre Bundles

2.1 Fibre Bundles

Definition 2.1.0.1 Let $\eta : E \rightarrow B$ be a bundle. Then, a **fiber bundle** is (η, E, B, F) with standard fiber F , such that for every general element $1 \rightarrow B$, the pullback of E along x is isomorphic to F .
i.e. $E_x \cong F$.

Chapter 3 Associated Bundles

Chapter 4 Vector Bundles